

Maintenance Manual — Pump P-204

1. Scope

This manual covers preventive and corrective maintenance for centrifugal pump P-204 (Kirloskar KDS-model, end-suction). It applies to reliability technicians and shift operators.

2. Nameplate

Tag : P-204
Duty : Boiler feed transfer
Bearing : DE 6309 / NDE 6209
Seal : Mechanical, single, balanced
PM interval: 1500 running hours (revised 2023)

For vibration limits and bearing criteria see Page 2.

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3. Vibration Limits (per ISO 10816)

Band	Velocity (mm/s)	Action
Normal	<= 4.5	Routine monitoring
Alert	4.5 - 7.1	Investigate, do not stop
Shutdown	>= 7.1	Trip and isolate

Continued operation above 5.5 mm/s in the alert band risks bearing damage and must be investigated the same shift.

4. Bearing Housing Temperature

Record bearing housing temperature at every reading. A delta above 15 C from the commissioning baseline indicates lubrication failure and precedes most bearing trips on P-204.

5. Bearing Inspection Checklist

- [] Housing temp delta vs baseline
- [] Grease colour / consistency
- [] Audible thud or grinding
- [] Axial / radial end play

6. Mechanical Seal Condition Matrix

Symptom	Likely cause	Action
Weeping at gland	Face wear	Schedule replace
Flush press drop	Clogged orifice	Clean flush line
Steam/vapour	Face lift-off	Stop, inspect

P-204 has a recorded history of seal-related trips at vibration levels in the 5.5 - 6.5 mm/s range. Inspect the seal before attributing high vibration solely to the bearing.

7. Coupling

Check coupling bolt torque (spec 45 Nm) during any high-vibration investigation. Loose coupling bolts mimic bearing symptoms.